

ZIDIO DEVELOPMENT

PROJECT FORESIGHT

Demand & Inventory Intelligence

Data Science & Analytics — Project Documentation & Reporting

Program	Zidio Development Internship
Project	FORESIGHT — Demand & Inventory Intelligence
Engagement	4-week / 1-month Data Science & Analytics project
Team	Team #8

Prepared from the project brief, Month 1 README, repository structure, and supplied project screenshots.

1. Overview

Project FORESIGHT is a Data Science & Analytics engagement focused on turning sales and inventory data into practical demand and inventory intelligence. The business problem is to reduce the operational impact of stockouts and overstock by using data-driven forecasting, risk identification and an accessible dashboard.

The Zidio project brief defines NorthBay Living as a direct-to-consumer home & lifestyle brand with approximately 200 active SKUs. The engagement is designed as a client-style project: analysis should be actionable for operations and finance stakeholders rather than remaining only inside notebooks.

Business Objective

- Develop an end-to-end analytics workflow from data ingestion through modelling and interactive visualisation.
- Generate SKU-level demand intelligence to support inventory planning.
- Use data analysis and machine learning to identify patterns that can inform operational decisions.
- Present outputs through an interactive Streamlit dashboard so that non-technical stakeholders can consume the insights.

Documented Month 1 Implementation

The supplied Month 1 README documents data ingestion and preprocessing, data cleaning and transformation, exploratory data analysis (EDA), feature engineering, machine-learning model development, Random Forest Regression, Matplotlib/Seaborn visualisation, an interactive Streamlit dashboard, and business-oriented data analysis.

Area	Documented status	Evidence in supplied material
Data ingestion & preparation	Implemented	Month 1 README / project features
EDA & feature engineering	Implemented	Month 1 README / project features
Random Forest Regression	Implemented	README reports trained model
Model metric	MAE: 3.48 reported	README status line
Interactive dashboard	Deployed / documented	README status line

Reporting note: the engagement brief specifies WAPE as the primary forecasting metric and requires fair baseline comparison through rolling-origin backtesting. The supplied screenshots confirm an MAE value of 3.48, but they do not provide a baseline/WAPE comparison or sufficient evidence to claim those acceptance criteria are complete. This report therefore preserves the documented result without inventing additional performance claims.

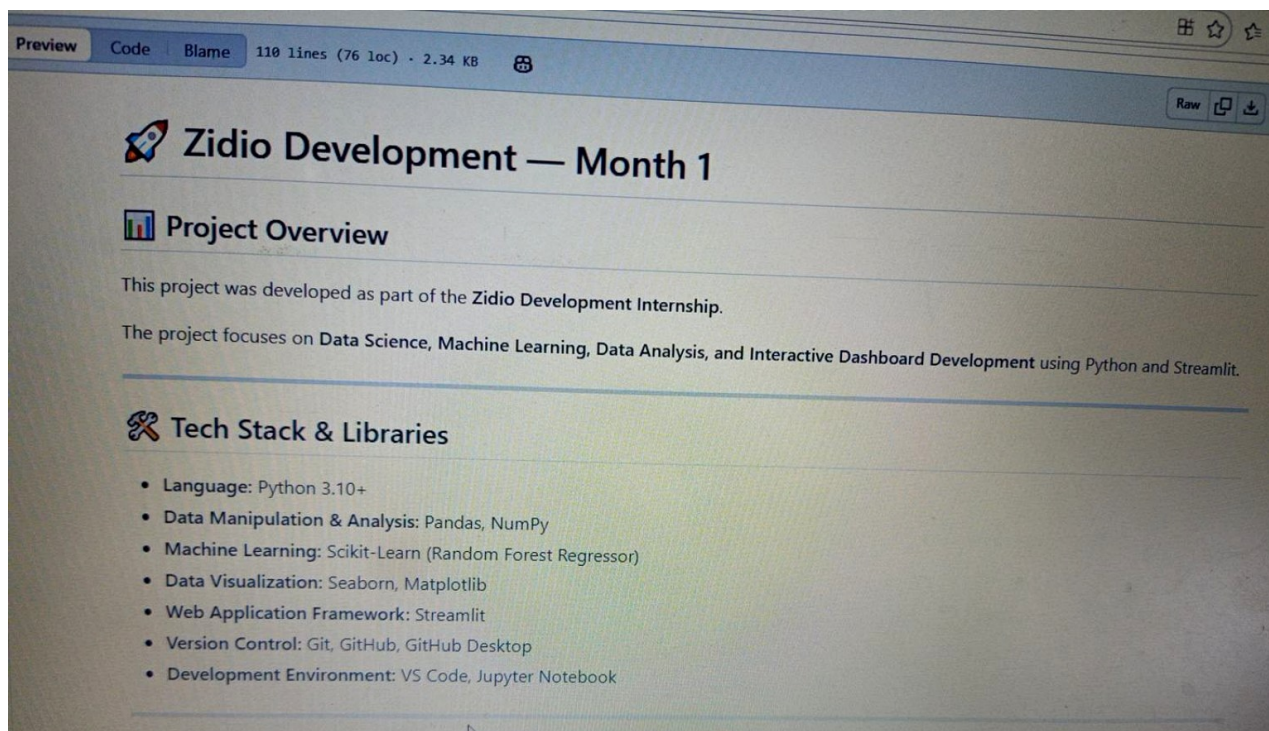


Figure 1. Month 1 README — project overview and documented technology stack.

2. Tech Stack

The project uses a practical Python-based analytics stack that supports data preparation, machine learning, visualisation, dashboarding, development and version control.

Layer	Technology	Purpose in the project
Programming	Python 3.10+	Core language for analytics, modelling and application code.
Data manipulation	Pandas, NumPy	Data ingestion, cleaning, transformation and numerical operations.
Machine learning	Scikit-learn — Random Forest Regression	Demand modelling / regression workflow.
Visualisation	Matplotlib, Seaborn	Exploratory analysis and analytical visualisations.
Dashboard	Streamlit	Interactive stakeholder-facing web dashboard.
Development	Jupyter Notebook, VS Code	Exploration, analysis and application development.
Version control	Git, GitHub, GitHub Desktop	Source control, collaboration and project traceability.

Repository & Reproducibility

The Month 1 repository is organised into application, data, notebooks and source/dependency areas. The README also provides local execution instructions: clone the repository, move into the Month 1 directory, install the requirements, and launch the Streamlit dashboard.

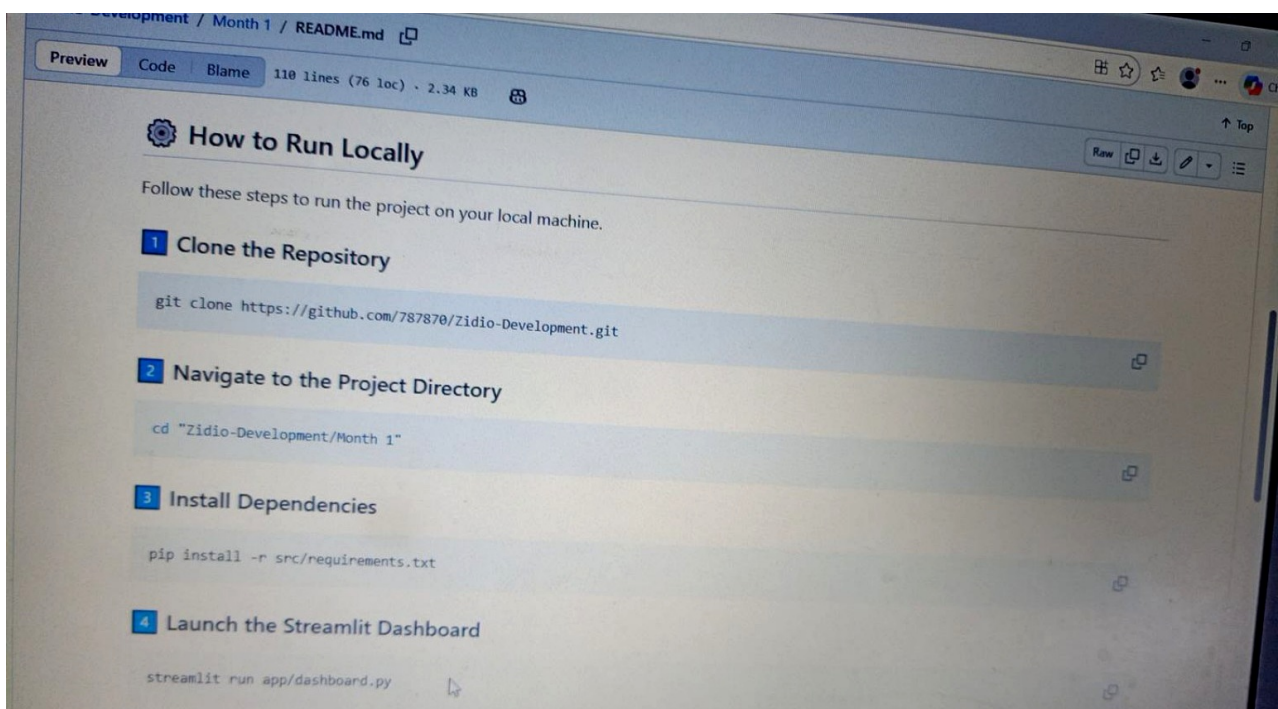


Figure 2. README — local execution workflow for the Month 1 project.

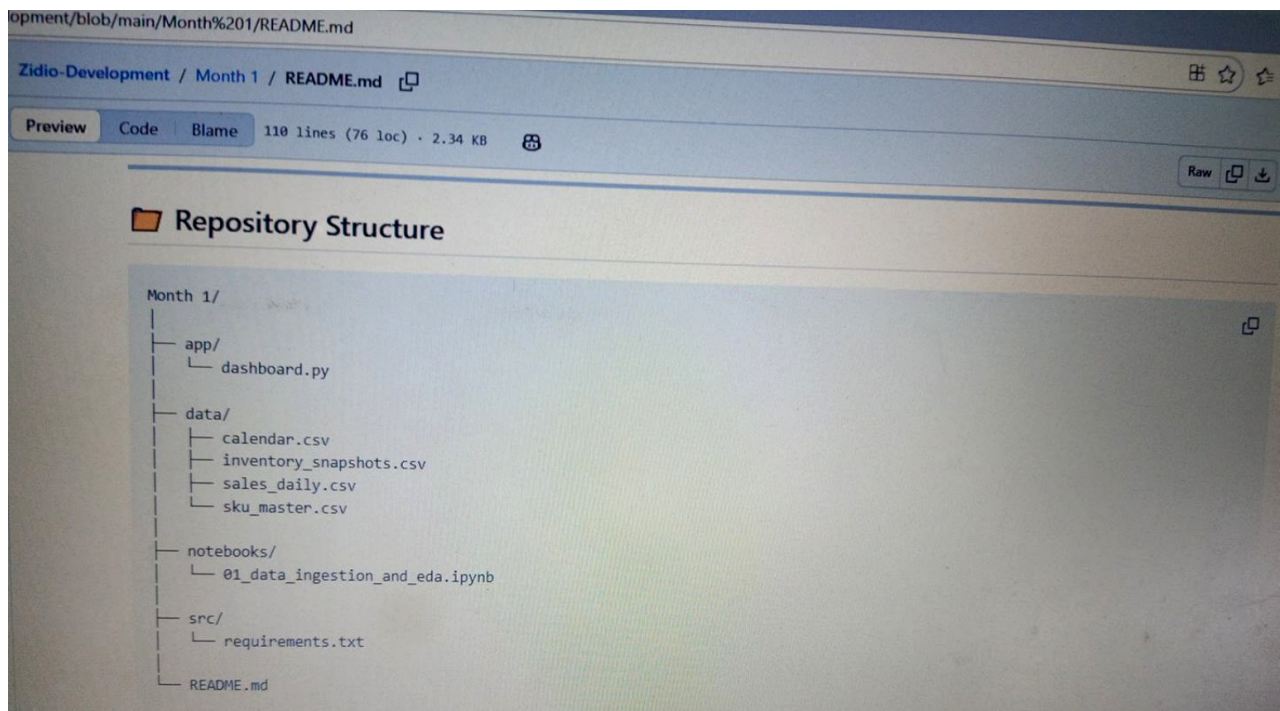
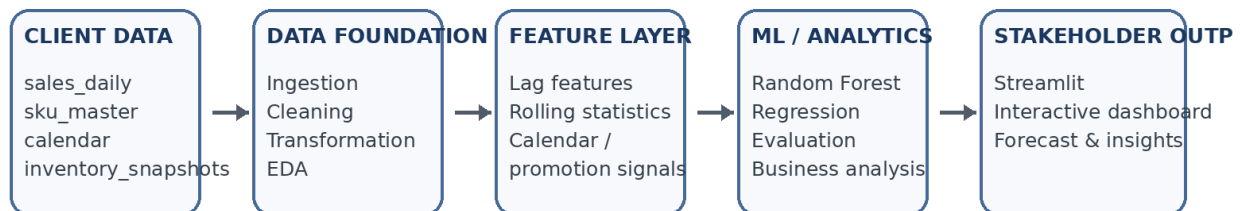


Figure 3. README — repository structure showing app, data, notebooks, src and requirements.

3. Architecture / Screenshots

The architecture follows a clear analytics flow: source extracts are ingested and cleaned, transformed into analysis-ready data, enriched through feature engineering, passed to the machine-learning layer, and surfaced through stakeholder-facing visualisation.

FORESIGHT — End-to-End Analytics Architecture



Reproducible workflow: data → preparation → features → model → visualisation → decision support

Figure 4. Project architecture reconstructed from the supplied README and Zidio engagement brief.

Data Layer

The project brief defines four core datasets: sales_daily (SKU-day sales, revenue, price and promotion flag), sku_master (product metadata and pricing), calendar (time and seasonal attributes), and inventory_snapshots (on-hand, on-order, lead time and reorder point). These datasets form the foundation for demand and inventory analysis.

Analytics & Modelling Layer

The documented Month 1 workflow covers ingestion, cleaning, EDA and feature engineering before machine-learning development. The supplied README specifically identifies Random Forest Regression as the implemented modelling approach.

Stakeholder Layer

The Streamlit dashboard is the principal documented interface. Its purpose is to move the work beyond notebooks and make analytical outputs easier to inspect and use.

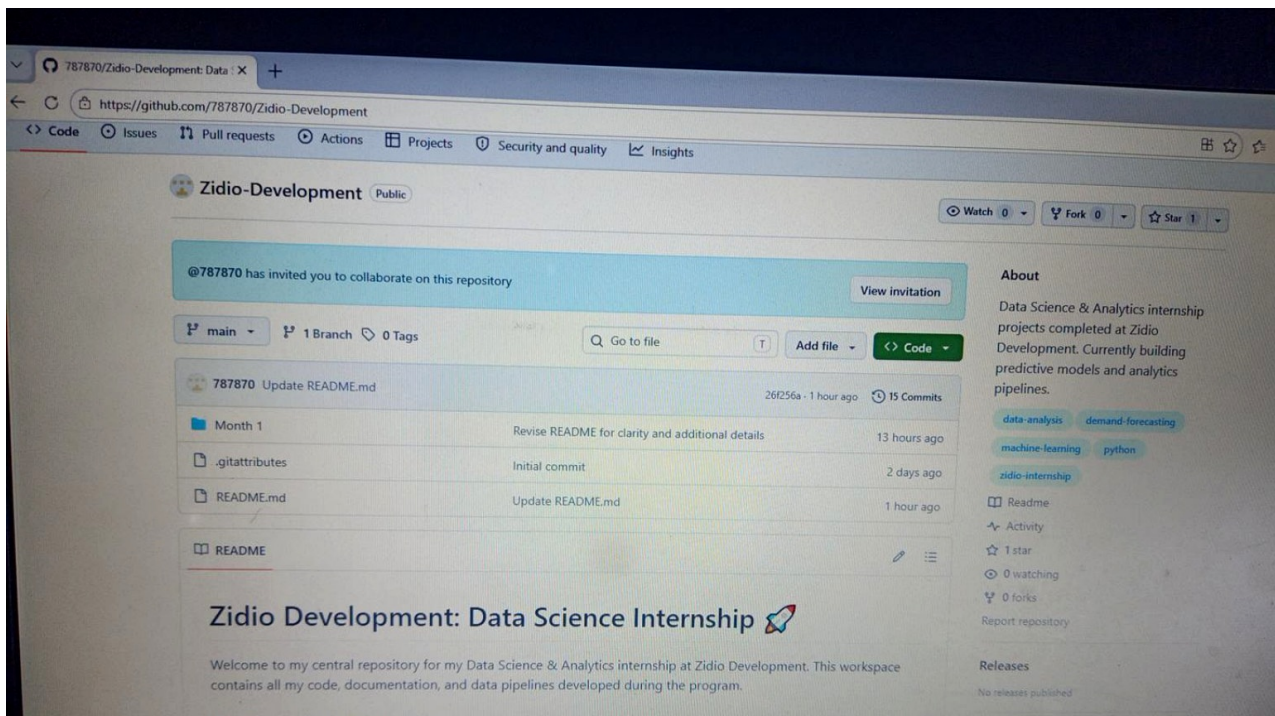


Figure 5. GitHub repository view showing the Month 1 project and README.

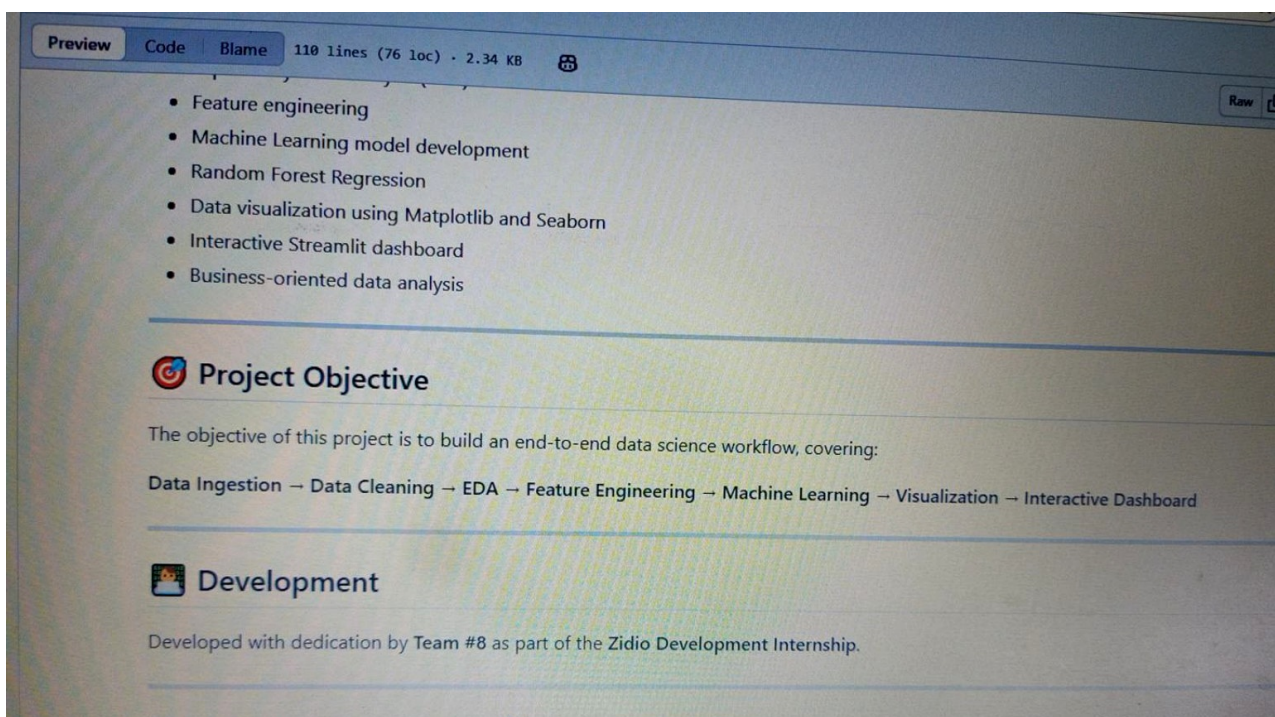


Figure 6. README — project features and end-to-end project objective.

Professional Reporting Check

- The screenshots are used as evidence of the repository, README, workflow and documented implementation.
- Only the model result explicitly visible in the supplied material (MAE 3.48) is reported as a measured result.
- No unverified WAPE, baseline win, risk-score performance, business savings or production-service result is fabricated.
- The architecture separates data preparation, modelling and stakeholder output, making the workflow easier to explain and reproduce.

4. Conclusion

Project FORESIGHT demonstrates an end-to-end Data Science & Analytics workflow for demand and inventory intelligence. The supplied Month 1 material shows a structured progression from data ingestion and preparation to EDA, feature engineering, Random Forest Regression, visualisation and an interactive Streamlit dashboard.

From a reporting perspective, the strongest feature of the work is that the technical components are connected to a business use case: understanding demand and inventory conditions so that operational teams can make better-informed stocking decisions.

Key Takeaways

- A reproducible Python analytics foundation has been established.
- The repository structure supports separation of application code, data, notebooks and dependencies.
- Random Forest Regression is documented as the Month 1 machine-learning approach, with MAE 3.48 reported in the README.
- Streamlit provides a practical interface for communicating analytical outputs to non-technical users.
- For the final client-grade submission, the documented brief should be used as the acceptance checklist—especially WAPE versus a seasonal-naïve baseline, rolling-origin backtesting, transparent risk scoring, and deployment evidence.

End of Project Documentation